

TrES-3b

R-2725

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160620 · created 2026-09-01T23:31:47+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457559.7881 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.048
Transit depth (Rp/Rs)^2	2.58 +/- 1.53 [%]
Orbital Inclination (inc)	86.61 +/- 2.33
Airmass coefficient 1 (a1)	0.986 +/- 0.042
Airmass coefficient 2 (a2)	0.01 +/- 0.13
Scatter in the residuals of the lightcurve fit is	4.26 %
Best Comparison Star	#2 - [80, 293]
Optimal Aperture	3.08
Optimal Annulus	13.94
Transit Duration (day)	0.0777 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.161 $\pm$ 0.048 vs 0.16309 (deviation 0.04 σ)
Duration fit vs published	0.0777 d vs 0.0567 d (deviation 2.31 σ)
Mid-time O-C	-3.9 min
Depth SNR	3.3
Residual scatter	4.26 %

Light curve

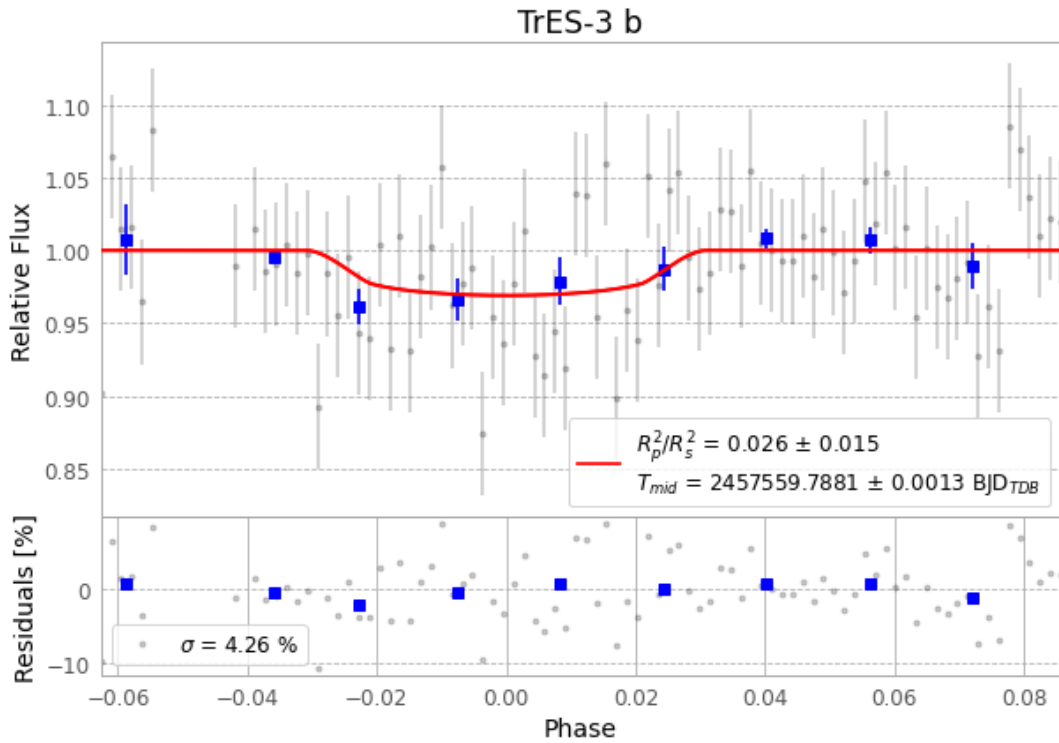


Figure 1 — FinalLightCurve_TrES-3 b_2016-06-19.png (SHA-256 cfe9141b225751b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [150, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9af0ffe8b7f23b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

94 FITS frames (62 MB), TRES-3160620045112.FITS → TRES-3160620093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160620.log (e310901ba18f...), FinalLightCurve_TrES-3 b_2016-06-19.png (cfe9141b2257...), FinalLightCurve_TrES-3 b_2016-06-19.csv (1cc5998fbb95...)

Compute resources

Wall time 212.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)